



House-sized asteroid near miss

On Oct 12, a house-sized asteroid passed about 42,000 km away from Earth. A whisker in space terms!
<http://digit.in/AstrdHouse>



Galaxy far, far away

Using a jet of radio waves, astronomers have begun to map the other side of the Milky Way.
<http://digit.in/milkywmap>

HOW WE TESTED

SETUP AND TESTING RIG

Every web browser was freshly installed on the system, and was verified to be updated to the latest version. During the tests, no extensions or add-ons were installed in the browser. To ensure only the web browser was majorly using the system resources, almost all the background applications were closed. This included anti-virus software. However, the antivirus was left running in our real-world tests. The tests were run on a single tab for the synthetic benchmarks. Power options were switched to High Performance. The synthetic benchmarks and real-world tests were run on a desktop PC with the following configuration:

- CPU: Intel Core i7-3770K
- GPU: MSI Gaming X Radeon RX 470
- RAM: Kingston HyperX DDR3 8GB
- Primary storage: SanDisk Extreme II
- OS: Windows 10 Pro

The entire testing was performed on an ethernet connection with a download and upload speed of 60 Mbps.

PERFORMANCE TEST

The performance test has been divided into synthetic benchmarks and real-world tests. Synthetic benchmarks will ensure there's absolute objectivity to the scores obtained. Real world tests will reflect more of a daily usage scenario that has been converted into scores.

To test how well does the web browser handle JavaScript, we used JetStream 1.1. Since Google's Octane is now unsupported, we chose not to use it. Also, SunSpider is no longer being maintained by the devs and they themselves recommend to use JetStream. Next, we used Speedometer

1.0 to test the browser's capability of handling web applications. Speedometer 2.0 is still under development, and we will use that when the final version is out. The HTML5 test judges whether the web browser supports HTML5 standards and elements. MotionMark and the Unity 3D benchmark 2015 test the browser's capability of supporting motion graphics. Finally, RoboHornet gives us an overall score of how certain essential aspects such as tables, DOM, CSS selectors, etc., affect applications like Google Spreadsheets, Presentations, etc. Essentially, the score will tell us if the browser is good enough for work-related use-cases.

The real-world testing is meant to bring some perspective with the scores so that readers can relate and understand how the browsers will fare in daily usage. In order to measure RAM usage, we let the system sit idle for a few minutes until it settled down. We created a batch file that launched 20 selected web sites at once when opened. Keeping a close watch on the usage, we recorded the RAM usage from the task manager once the webpages finished loading. To verify webpage load times, we ran a JavaScript query saved in an HTML file locally that calculated the time taken for a webpage to load, in our case, the Digit website. The initial load time was recorded and then the same page was loaded five times to get an average score.

FEATURES EVALUATION

Web browsers have an exhaustive list of features and listing down each one of them would be practically impossible here. A browser should support all the basic features required to seamlessly browse

websites and enhance the user's browsing experience to an extent. We do love web browsers that offer a lot of features, but not at the cost of performance. Speaking of extra features, most of them can be accomplished by using extensions or as Firefox likes to call them, add-ons. Some browsers have gone the extra mile to include few of them as a native feature. Both have been evaluated where native features has been rewarded slightly more than extension-enabled support. This is because of less effort and more importantly, you can trust the browser's implementation rather than downloading a potentially harmful extension. The difference in weights is kept low in decimal points so that ultimately, the final score doesn't take a hit for being a native feature rather than an extension-enabled one.

VERDICT

Because of the absence of the build quality and price parameters in this comparison test, the final verdict will be distributed in three separate awards.

To determine the final winner, an overall score will be calculated with a weighted sum of both performance and features. The recipient of the Best Web Browser award will be deemed capable of offering the best balance in performance and features.

The Best Performer award will be given to the web browser with the highest score in both synthetic benchmarks and real-world testing. This award will reflect only performance and no other aspect.

The Editor's Pick award will be given to the Editor's favourite web browser based on the findings of this test.

desktop system with a healthy amount of RAM and powerful CPU. We can't guarantee the same performance will be achieved on laptops. Chrome also had the lowest page load times which means that webpages will load the fastest here.

Extension support is massive on Chrome. Name a feature you would like in the browser and an extension might just be waiting for you already. Because of the easy availability of extensions, some convenient features aren't native to Chrome – like a screen-

shot tool. Customisability in terms of look is also limited here. If you want a hassle-free, fast and stable browser, you should be using Chrome, and you probably already are.

INTERNET EXPLORER

Internet Explorer (IE) became king of the browsers by knocking Netscape off its throne many years ago. However, it did it by being bundled in Windows as the default browser, and got a lot of criticism for it. The roles are reversed now, and IE is going to be phased out

slowly as Windows is pushing its new Edge browser to its customers. IE scored the lowest in every synthetic benchmark except RoboHornet – a test that suggests IE might actually be decent for working with tables in, say, Google Spreadsheets. WebGL support is poor on Internet Explorer, and the MotionMark test froze in the third render screen and refused to progress. The Unity 3D WebGL 2015 benchmark struggled a lot with poor frame rates, and IE also couldn't finish the JetStream test. After opening a mere 10